



Triangle-Free Graphs with Diameter 2

Alice Devillers, Nina Kamčev, Brendan D. McKay, Padraig Ó Catháin, Gordon Royle, Geertrui Van de Voorde, Ian M. Wanless, David R. Wood

Abstract The class of graphs with diameter 2 and girth 5 is finite (up to isomorphism). What if the girth 5 assumption is relaxed? Apart from stars, are there infinitely many triangle-free graphs with diameter 2 and no $K_{2,3}$ subgraph? This question is related to the existence of triangle-free strongly regular graphs, but allowing for a range of co-degrees gives the question a more extremal flavour. More generally, for fixed s and t , are there infinitely many twin-free triangle-free $K_{s,t}$ -free graphs with diameter 2? This paper presents partial results regarding these questions, including computational results, and potential Cayley-graph and probabilistic constructions.

Alice Devillers and Gordon Royle

Department of Mathematics and Statistics, University of Western Australia, Perth, Australia

e-mail: alice.devillers,gordon.royle@uwa.edu.au

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Nina Kamčev

Department of Mathematics, Faculty of Science, University of Zagreb, Croatia

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Brendan D. McKay

School of Computing, Australian National University, Canberra, Australia

e-mail: Brendan.McKay@anu.edu.au

Padraig Ó Catháin

Fiontar agus Scoil na Gaeilge, Dublin City University, Dublin, Ireland

e-mail: padraig.ocathain@dcu.ie

Geertrui Van de Voorde

School of Mathematics and Statistics, University of Canterbury, Christchurch, New Zealand

e-mail: geertrui.vandevoorde@canterbury.ac.nz

Ian M. Wanless and David R. Wood

School of Mathematics, Monash University, Melbourne, Australia

e-mail: ian.wanless,david.wood@monash.edu

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1 Introduction

The class of triangle-free graphs with diameter 2 is surprisingly rich. For example, the Kneser¹ graph $K(3k-1, k)$ is triangle-free with diameter 2, which Lovász [16] famously proved has chromatic number $k+1$. However, if we impose some other properties on the class, then the diversity of examples quickly drops. For example, consider such graphs with girth 5. In one of the most striking achievements of algebraic graph theory, Hoffman and Singleton [14] proved that every graph with diameter 2 and girth 5 is k -regular with k^2+1 vertices, where $k \in \{2, 3, 7, 57\}$ (see [13]). The only 2-regular example is the 5-cycle, the only 3-regular example is the Petersen graph on 10 vertices, the only 7-regular example is the Hoffman-Singleton graph on 50 vertices, and it is open whether there exists a 57-regular graph with diameter 2 and girth 5, which would have 3250 vertices. Put another way (since every tree with diameter 2 is a star and $K_{2,2} \cong C_4$), the only triangle-free $K_{2,2}$ -free graph with diameter 2 and with $n \geq 3251$ vertices is the star graph $K_{1,n-1}$. The following conjecture replaces $K_{2,2}$ in this result by $K_{2,t}$:

Conjecture 1. For every integer $t \geq 2$, there exists n_0 such that if G is a triangle-free $K_{2,t}$ -free graph G with diameter 2 and with $n \geq n_0$ vertices, then G is the star graph $K_{1,n-1}$.

This conjecture was posed by the final author at the “Extremal Problems in Graphs, Designs, and Geometries” workshop at the Mathematical Research Institute MATRIX. This paper reports on work related to [Conjecture 1](#) completed during and after the workshop. [Section 2](#) presents results about the degrees of vertices in n -vertex graphs that arise in [Conjecture 1](#). In particular, for fixed t , we show that the minimum degree, average degree and maximum degree are all within a constant factor of $\sqrt{n}$. The next two sections focus on the $t = 3$ case of [Conjecture 1](#), which is the first open case. We use computer search to find examples of $K_{2,3}$ -free graphs with diameter 2. In all, we have found 5,936,056 examples, but nothing suggestive of an infinite class. [Section 4.1](#) explores relationships between the $t = 3$ case of [Conjecture 1](#) and questions about strongly regular graphs. [Section 4.2](#) investigates potential Cayley graphs satisfying [Conjecture 1](#), and shows that groups of order 2^m satisfy the conjecture. [Section 5](#) considers a possible generalisation of [Conjecture 1](#) replacing $K_{2,t}$ by $K_{s,t}$. Finally, [Section 6](#) discusses whether random graphs provide counterexamples to [Conjecture 1](#) (and the generalisation presented in [Section 5](#)). The conjectures turn out to be interesting from this perspective as well.

To conclude this introduction, note that results of Plesník [20] and Chakraborty, Das, Mukherjee, Kant Sahoo, and Sen [7] respectively imply [Conjecture 1](#) for planar graphs and projective planar graphs, but this says nothing about the general question.

¹ The Kneser graph $K(n, k)$ has vertex-set all k -sets of a ground set of size n , where AB is an edge if and only if $A \cap B = \emptyset$.

2 Degree Considerations

As discussed above, the $t = 2$ case of [Conjecture 1](#) holds. The proof starts by showing that G is regular. The following observation generalises this property. Let $\delta(G)$ and $\Delta(G)$ be the minimum and maximum degree of G respectively.

Theorem 1. *For every integer $t \geq 2$, if G is a triangle-free $K_{2,t}$ -free graph with diameter 2 and G is not a star graph, then*

$$\Delta(G) \leq (t-1)((t-1)\delta(G) - t + 2) - t + 2.$$

Proof. Star graphs are the only triangle-free graphs G with diameter 2 and $\delta(G) = 1$. So we may assume that $\delta(G) \geq 2$.

First suppose that G is bipartite with bipartition $\{A, B\}$. If there exist non-adjacent vertices $v \in A$ and $w \in B$, then $\text{dist}_G(v, w) \geq 3$. So G is a complete bipartite graph. Since G is not a star graph, $G \cong K_{a,b}$ for some integers a, b with $b \geq a \geq 2$. Since G is $K_{2,t}$ -free, $\Delta(G) = b \leq t-1$, which is at most $(t-1)((t-1)\delta(G) - t + 2) - t + 2$ since $\delta(G) \geq 2$.

This completes the proof in the bipartite case. Now assume that G is not bipartite. Our goal is to show that $\deg(v) \leq (t-1)((t-1)\deg(w) - t + 2) - t + 2$ for any vertices v, w of G .

First consider vertices v, w at distance 2 in G . Let $C := N_G(v) \cap N_G(w)$. Thus $|C| \in \{1, \dots, t-1\}$. Let $A := N_G(v) \setminus C$ and $B := N_G(w) \setminus C$. Since G is triangle-free, A and B are independent sets. Let H be the bipartite subgraph $G[A \cup B]$. Consider a vertex $x \in A$. Thus x is not adjacent to w (since $x \notin C$), and there is no neighbour of x in C (since G is triangle-free). Hence x has a neighbour in B (since $\text{dist}_G(x, w) = 2$). On the other hand, each vertex in B has at most $t-1$ neighbours in A . Hence $|A| \leq |E(H)| \leq (t-1)|B|$. Now,

$$\begin{aligned} \deg(v) &= |C| + |A| \leq |C| + (t-1)|B| = |C| + (t-1)(\deg(w) - |C|) \\ &= (t-1)\deg(w) - (t-2)|C| \\ &\leq (t-1)\deg(w) - t + 2. \end{aligned}$$

Now consider an edge vw of G . Suppose there is a vertex x of G adjacent to neither v nor w . Thus $\text{dist}_G(x, v) = 2$ and $\text{dist}_G(x, w) = 2$. As proved above, $\deg(v) \leq (t-1)\deg(x) - t + 2$ and $\deg(x) \leq (t-1)\deg(w) - t + 2$. Thus $\deg(v) \leq (t-1)((t-1)\deg(w) - t + 2) - t + 2$, as desired.

Now assume that each vertex $x \in V(G) \setminus \{v, w\}$ is adjacent to v or w . Since G is triangle-free, x is adjacent to exactly one of v and w . Also since G is triangle-free, $N_G(v) \setminus \{w\}$ is an independent set, and $N_G(w) \setminus \{v\}$ is an independent set. Thus G is bipartite with bipartition $\{N_G(v), N_G(w)\}$, which is a contradiction.

The following result describes the degree sequences that arise in [Conjecture 1](#).

Proposition 2 *For every integer $t \geq 2$, if G is a triangle-free n -vertex graph with diameter 2 and no $K_{2,t}$ subgraph, then*

$$\frac{1}{t-1} \sum_{v \in V(G)} \left(\deg(v)^2 + (t-2) \deg(v) \right) \leq n(n-1) \leq \sum_{v \in V(G)} \deg(v)^2.$$

Proof. Let $\mathcal{P}$ be the set $\{(v, \{u, w\}) : uv, vw \in E(G), u \neq w\}$. Then $|\mathcal{P}| = \sum_{v \in V(G)} \binom{\deg(v)}{2}$. Since G has diameter 2, for each non-adjacent pair $u, w \in V(G)$ there exists $v \in V(G)$ such that $(v, \{u, w\}) \in \mathcal{P}$. Thus,

$$\sum_{v \in V(G)} \binom{\deg(v)}{2} = |\mathcal{P}| \geq \binom{n}{2} - |E(G)| = \binom{n}{2} - \frac{1}{2} \sum_{v \in V(G)} \deg(v).$$

Hence,

$$\sum_{v \in V(G)} \deg(v)^2 \geq n(n-1).$$

This proves the second inequality. For the first inequality, charge each element $(v, \{u, w\}) \in \mathcal{P}$ to $\{u, w\}$. Since G is triangle-free, $uw \notin E(G)$. Since G has no $K_{2,t}$ subgraph, at most $t-1$ elements of $\mathcal{P}$ are charged to $\{u, w\}$, so.

$$\sum_{v \in V(G)} \binom{\deg(v)}{2} = |\mathcal{P}| \leq (t-1) \left(\binom{n}{2} - |E(G)| \right).$$

Thus,

$$\sum_{v \in V(G)} \left(\deg(v)^2 - \deg(v) \right) \leq (t-1) \left(n(n-1) - \sum_{v \in V(G)} \deg(v) \right).$$

Hence,

$$\sum_{v \in V(G)} \left(\deg(v)^2 + (t-2) \deg(v) \right) \leq (t-1) n(n-1),$$

which implies the first inequality.

Proposition 3 *For every integer $t \geq 2$, if G is an n -vertex triangle-free $K_{2,t}$ -free graph with diameter 2, and G is not a star graph, then G has minimum degree at least $(1 + o(1)) \frac{1}{(t-1)^2} \sqrt{n}$ and at most $(1 + o(1)) \sqrt{tn}$, and G has maximum degree at least $\sqrt{n-1}$.*

Proof. The Kővári–Sós–Turán Theorem [15] says that for any integers $t \geq s \geq 1$ there exists c such that any graph on n vertices with at least $cn^{2-1/s}$ edges contains $K_{s,t}$. For $s = 2$ this result is tight and even the best possible constant c is known. Füredi [11] showed that $\text{ex}(n, K_{2,t}) = (1 + o(1)) \frac{\sqrt{t}}{2} n^{3/2}$ (also see [12, Theorem 3.11]). So $|E(G)| \leq (1 + o(1)) \frac{\sqrt{t}}{2} n^{3/2}$ edges. That is, G has average degree at most $(1 + o(1)) \sqrt{tn}$, which implies the same bound on the minimum degree. On the other hand, since G has diameter 2, the maximum degree of G is at least $\sqrt{n-1}$, which implies the minimum degree is at least $(1 + o(1)) \frac{1}{(t-1)^2} \sqrt{n}$ by Theorem 1.

3 Examples

This section reports the findings of a computer search for graphs with the following properties:

- there are no triangles,
- each pair of non-adjacent vertices have exactly 1 or 2 common neighbours (that is, diameter 2 and $K_{2,3}$ -free), and
- the graph is not a star $K_{1,n-1}$.

[Conjecture 1](#) with $t = 3$ says there are finitely many such graphs. [Proposition 2](#) says that any such n -vertex graph G satisfies

$$\sum_{v \in V(G)} \binom{\deg(v) + 1}{2} \leq n(n - 1) \leq \sum_{v \in V(G)} \deg(v)^2.$$

If G is d -regular, then $1 + \binom{d+1}{2} \leq n \leq 1 + d^2$, with both extremes giving strongly regular graphs.

The following tables describe the 5,936,197 examples that we have found, the largest on 120 vertices. The fifth and sixth columns are the order and the number of orbits of the automorphism group, respectively.

vertices	edges	degrees	girth	group	orbits	notes
4	4	2 ⁴	4	8	1	4-cycle
5	5	2 ⁵	5	10	1	5-cycle
6	7	2 ⁴ 3 ²	4	4	3	subdivided $K_{2,3}$
7	9	2 ⁴ 3 ² 4	4	8	3	
8	12	3 ⁸	4	16	1	Möbius ladder
9	14	3 ⁸ 4	4	8	3	
10	15	3 ¹⁰	5	120	1	Petersen graph
10	17	3 ⁶ 4 ⁴	4	4	4	
11	19	3 ⁶ 4 ⁵	4	24	3	
11	20	3 ⁵ 4 ⁵ 5	4	10	3	Groetzsch graph
12	23	3 ³ 4 ⁸ 5	4	12	4	
12	24	4 ¹²	4	48	1	
13	24	3 ⁴ 4 ⁹	4	48	3	
13	26	4 ¹³	4	52	1	
13	27	3 4 ⁹ 5 ³	4	12	4	
14	31	4 ⁸ 5 ⁶	4	48	2	
15	35	4 ⁵ 5 ¹⁰	4	120	2	
16	34	4 ¹² 5 ⁴	4	32	3	
16	40	5 ¹⁶	4	1920	1	Clebsch graph
17	40	4 ⁵ 5 ¹²	4	48	3	
18	39	4 ¹² 5 ⁶	4	24	3	

vertices	edges	degrees	girth	group	orbits	notes
20	46	$4^{14} 6^6$	4	96	3	
20	50	5^{20}	4	320	1	
22	55	5^{22}	4	96	3	
22	57	$5^{18} 6^4$	4	1-4	6-22	(2 graphs)
23	61	$4 5^{14} 6^8$	4	4	9	
24	64	$5^{16} 6^8$	4	2-384	3-12	(9 graphs)
26	73	$5^{10} 6^{16}$	4	4	8	
26	75	$5^6 6^{20}$	4	20	3	
28	84	6^{28}	4	2-56	2-21	(6 graphs)
29	84	$5^7 6^{21} 7$	4	28	4	
31	93	$5^4 6^{24} 7^2 8$	4	4	11	
32	96	6^{32}	4	48-1920	1-3	(3 graphs)
32	102	$6^{20} 7^{12}$	4	2	16	
35	112	$6^{21} 7^{14}$	4	42	2	
36	119	$6^{21} 7^8 8^7$	4	21	4	
36	120	$6^{15} 7^{20} 10$	4	20	6	
36	122	$6^{10} 7^{24} 8^2$	4	5-10	6-12	(8 graphs)
36	122	$6^{12} 7^{20} 8^4$	4	1-3	14-36	(2 graphs)
36	123	$6^9 7^{24} 8^3$	4	1-2	20-36	(3 graphs)
36	124	$6^6 7^{28} 8^2$	4	1-2	20-36	(4 graphs)
36	125	$6^3 7^{32} 8$	4	1-6	9-36	(3 graphs)
36	126	7^{36}	4	7	6	
36	128	$6^4 7^{24} 8^8$	4	1-2	19-36	(5 graphs)
37	127	$6^{21} 8^{16}$	4	42	3	
37	130	$6^{12} 7^{12} 8^{13}$	4	1-3	15-37	(2 graphs)
37	131	$6^9 7^{16} 8^{12}$	4	1-2	21-37	(3 graphs)
37	132	$6^6 7^{20} 8^{11}$	4	1-2	21-37	(4 graphs)
37	133	$6^3 7^{24} 8^{10}$	4	1-6	10-37	(3 graphs)
37	134	$7^{28} 8^9$	4	7	7	
39	143	$7^{26} 8^{13}$	4	1-78	2-39	(536 graphs)
40	140	7^{40}	4	1-40	2-40	(19 graphs)
40	144	$7^{32} 8^8$	4	8-32	4-7	(4 graphs)
45	180	8^{45}	4	30-1440	1-2	(2 graphs)
46	169	$7^{34} 8^{10} 10^2$	4	1-4	14-46	(3 graphs)
46	170	$7^{30} 8^{15} 10$	4	1-10	8-46	(6 graphs)
46	189	$8^{36} 9^{10}$	4	144	3	
46	195	$7^3 8^{21} 9^{19} 10^3$	4	6	10	
47	174	$7^{34} 8^8 9^4 10$	4	1-4	14-47	(6 graphs)
47	175	$7^{30} 8^{13} 9^4$	4	1-2	27-47	(10 graphs)
47	198	$8^{28} 9^{18} 10$	4	32	6	
48	172	$7^{46} 11^2$	4	40	6	
48	174	$7^{41} 8^4 9^2 11$	4	2	26-28	(3 graphs)

vertices	edges	degrees	girth	group	orbits	notes
48	176	$7^{36} 8^8 9^4$	4	1–8	10–48	(31 graphs)
48	192	8^{48}	4	1–240	2–48	(404 graphs)
48	207	$8^{21} 9^{24} 10^3$	4	12	9	
49	174	$7^{48} 12$	4	1440	3	
49	177	$7^{42} 8^4 9^2 10$	4	8	10	
49	216	$8^{15} 9^{28} 10^6$	4	8–48	5–11	(2 graphs)
49	219	$8^8 9^{36} 10^5$	4	8	11	
50	175	7^{50}	5	252000	1	Hoffman–Singleton graph
50	217	$7^8 9^{42}$	4	336	3	
50	225	$8^{10} 9^{30} 10^{10}$	4	8–20	5–13	(2 graphs)
50	227	$8^6 9^{34} 10^{10}$	4	4	16	
51	230	$8^{10} 9^{30} 10^{11}$	4	60	4	
51	234	$8^6 9^{30} 10^{15}$	4	8–48	5–12	(2 graphs)
51	235	$8^5 9^{30} 10^{16}$	4	10	8	
52	243	$8^3 9^{28} 10^{21}$	4	12	10	
52	244	$9^{32} 10^{20}$	4	128	3	
53	252	$8 9^{24} 10^{28}$	4	32	6	
54	261	$9^{18} 10^{36}$	4	288	2	
55	270	$9^{10} 10^{45}$	4	1440	2	
56	252	9^{56}	4	4–28	2–14	(19 graphs)
56	280	10^{56}	4	80640	1	Gewirtz graph
58	267	$8^2 9^{42} 10^{14}$	4	2–14	5–29	(20 graphs)
60	270	9^{60}	4	30–120	1–2	(2 graphs)
60	284	$9^{32} 10^{28}$	4	4–224	2–15	(28 graphs)
62	301	$9^{18} 10^{44}$	4	16–112	4–10	(20 graphs)
64	320	10^{64}	4	32–1792	2–9	(9 graphs)
70	315	9^{70}	4	70	2	
72	360	10^{72}	4	36–144	1–2	(2 graphs)
80	400	10^{80}	4	70–80	2–4	(2 graphs)
90	495	11^{90}	4	20–40	3–6	(3 graphs)
95	545	$11^{50} 12^{45}$	4	100	4	(2 graphs)
100	600	12^{100}	4	1000	2	
120	780	13^{120}	4	3840	1	
120	780	13^{120}	4	≥ 2	≥ 2	(5,934,946 graphs)

The graphs in graph6, Magma or adjacency matrix format, except for the non-vertex-transitive graphs on 120 vertices can be downloaded from

<https://users.cecs.anu.edu.au/~bdm/data/woodgraphs.g6>

<https://users.cecs.anu.edu.au/~bdm/data/woodgraphs.magma>

<https://users.cecs.anu.edu.au/~bdm/data/woodgraphs.am>

The 53733 known graphs on 120 vertices with at most 12 orbits are at

<https://users.cecs.anu.edu.au/~bdm/data/woodgraphs120o12.g6>

The collection is complete for the following cases:

- everything up to 19 vertices,
- 20 vertices and minimum degree at least 5 (any others have minimum degree 4),
- 22 vertices and regular,
- 23 vertices and regular,
- 24 vertices and regular of degree 6 (degree 5 is still possible),
- vertex-transitive graphs up to 49 vertices,
- Cayley graphs of all groups up to order 320 except 256 and one group of order 320,
- vertex-transitive graphs up to 255 vertices whose automorphism group has a transitive subgroup of order twice the number of vertices,
- vertex-transitive graphs up to 170 vertices whose automorphism group has a transitive subgroup of order three times the number of vertices,
- graphs up to order 158 with a group acting regularly in parallel on two orbits.
- graphs of order n with a group of order n acting in parallel on two orbits, for $n \leq 428$,
- graphs of order n with a group of order $3n/2$ acting in parallel on two orbits, for $n \leq 336$.

3.1 Summary of the data

The computational results of the previous subsection show that there are large numbers of small triangle-free $K_{2,3}$ -free graphs of diameter 2. It is natural to examine these to determine if there are any obvious patterns that might be extended or generalised to an infinite family.

Indeed, there are several distinct “clusters” of graphs that collectively contain the vast majority of the examples. In particular, there is a cluster of 536 graphs on 39 vertices, one of 390 graphs on 48 vertices and one of nearly 6 million graphs on 120 vertices. Although we can describe each of these clusters quite precisely, none of them seem to generalise to an infinite family.

The 4-cycle and 5-cycle constructions:

Many of the graphs in the list of examples can be constructed in the same general fashion. Let $C_1, C_2, \dots, C_k$ and $D_1, D_2, \dots, D_\ell$ be a collection of $k + \ell$ vertex-disjoint 4-cycles, and form a graph by adding a perfect matching between each C_i and D_j in such a way that the subgraph induced by $V(C_i \cup D_j)$ is the cubic Möbius ladder on 8 vertices (also known as the Wagner graph V_8). Equivalently, we start with a complete bipartite graph $K_{k,\ell}$ and then expand each vertex to a 4-cycle and each edge to a 4-edge matching. This construction produces all but a handful of the examples when the number of vertices is a multiple of 4; for example, all but 11 of the 391 graphs on 48 vertices arise from the 4-cycle construction. An analogous construction

uses 5-cycles rather than 4-cycles in such a way that the subgraph induced by each connected pair of 5-cycles is the Petersen graph. A Ramsey Theory argument shows that neither of these constructions can produce an infinite family.

Edge rotations of 39-vertex graphs:

Define an *edge rotation* of a graph G to be the graph $G - uv + uw$ obtained by deleting an edge uv and adding a non-edge uw . Occasionally this operation preserves the property of being a triangle-free $K_{2,3}$ -free graph of diameter two and can thus be used to construct new examples from old. This is particularly effective on 39 vertices—every graph in the cluster of 536 graphs can be obtained from any other by a sequence of edge rotations where every intermediate graph is a triangle-free $K_{2,3}$ -free graph of diameter two.

120-vertex double covers:

One example with 120 vertices and degree 13 was found with a vertex-transitive group of order 3840, and we noticed that it was a double-cover of a vertex-transitive graph of order 60. Define the following operation on 120-vertex graphs: take a fixed-point-free involution and identify the two vertices in each orbit without removing loops or multiple edges. Then expand the resulting 60-vertex multigraph in all possible ways to 120 vertices using the inverse operation. Under circumstances which are easy to teach to a SAT-solver, this sometimes produces additional 120-vertex triangle-free $K_{2,3}$ -free graphs of diameter two. Then we can repeat the process until no further such graphs are produced. After many iterations, a total of 5,934,947 graphs of order 120 were found. Only the initial graph is vertex-transitive.

Miscellaneous constructions:

Any induced subgraph H of a triangle-free $K_{2,3}$ -free graph of diameter 2 inherits the properties of being triangle-free and $K_{2,3}$ free. If it happens to also have diameter 2, then H is another example.

A *Ryser-switch* on a graph G is a degree-preserving graph operation that removes two existing edges uv , wx and inserts two new edges uw and vx (where $\{u, v, w, x\}$ are distinct, $uv, wx \in E(G)$ and $uw, vx \notin E(G)$). Occasionally a new triangle-free $K_{2,3}$ -free graph of diameter 2 can be obtained from an existing one.

The list of examples presented in the previous section is closed under taking induced subgraphs and Ryser switches.

4 Symmetric Constructions

It is natural to consider symmetric graphs to be possible counterexamples to [Conjecture 1](#). We first consider strongly regular graphs, and then consider Cayley graphs.

4.1 Strongly regular graphs

For integers $\lambda \geq 0$ and $\mu \geq 1$, a k -regular graph G on n vertices is (λ, μ) -*strongly regular* if every pair of adjacent vertices have λ common neighbours, and every pair of distinct non-adjacent vertices have μ common neighbours. Our notation for such a graph is $\text{srg}(n, k, \lambda, \mu)$. A graph is *strongly regular* if it is (λ, μ) -strongly regular for some integers $\lambda \geq 0$ and $\mu \geq 1$.

The case $\lambda = 0$ corresponds to triangle-free strongly regular graphs. Complete bipartite graphs $K_{n,n}$ are triangle-free strongly regular graphs, said to be *trivial*. There are seven known non-trivial triangle-free strongly regular graphs (the 5-cycle, the Petersen graph, the Clebsch graph, the Hoffman–Singleton graph, the Gewirtz graph, the Mesner-M22 graph, and the Higman–Sims graph). It is open whether there are infinitely many non-trivial triangle-free strongly regular graphs. Biggs [1] showed that for any positive integer $\mu \notin \{2, 4, 6\}$ there are finitely many triangle-free $(0, \mu)$ -strongly regular graphs; see [2–4, 9] for related results.

Suppose that for some $\mu \in \{2, 4, 6\}$, there is an infinite family $\mathcal{G}$ of non-trivial triangle-free $(0, \mu)$ -strongly regular graphs. Since $\mu \geq 1$, every graph in $\mathcal{G}$ has diameter 2. By assumption, every graph in $\mathcal{G}$ is $K_{2, \mu+1}$ -free. Since every graph in $\mathcal{G}$ is regular and non-trivial, no graph in $\mathcal{G}$ is a star. Thus, [Conjecture 1](#) fails with $t = \mu + 1$. Hence, [Conjecture 1](#) with $t = 7$ implies there are finitely many non-trivial triangle-free strongly regular graphs. It is interesting that [Conjecture 1](#) replaces the algebraic setting for the study of strongly regular graphs by a more extremal viewpoint.

4.2 Cayley graphs on abelian groups

Let A be a finite abelian group written additively, and S an inverse-closed subset of A . The *Cayley graph* (A, S) is defined to be the graph with vertex set A , where two vertices $x, y \in A$ are adjacent if and only if $x - y \in S$. Define $T := \{x + x : x \in S\}$.

Lemma 1. *Let (A, S) be any Cayley graph.*

- (a) *The number of common neighbours of two non-adjacent vertices in (A, S) is even, unless their difference is in T .*
- (b) *(A, S) is K_3 -free if and only if there is no set of elements $x, y, z \in S$ such that $x + y = z$.*

- (c) (A, S) has diameter 2 if and only if for every $z \notin \{S \cup \{0\}\}$ there exist $x, y \in S$ such that $x + y = z$.
- (d) (A, S) is $K_{2,3}$ -free if and only if the following hold. For every $z \notin \{T \cup \{0\}\}$ there exists at most one pair $x, y \in S$ such that $x + y = z$. For each $z \in T$ such that $z \neq 0$, if $x + y = z$ then $x = y$; and the number of $x \in S$ for which $x + x = z$ is at most two.

Proof. (a) Let x, y be non-adjacent vertices in (A, S) . If x, y have no common neighbours, then the claim holds. Otherwise, let z be a common neighbour of x, y . Then there exist $s_1, s_2 \in S$ such that $z = x + s_1$ and $y = z + s_2$. If $s_1 = s_2$ then $x - y \in T$. Otherwise, $z' = x + s_2$ is a second common neighbour of x and y . Hence, the common neighbours of x, y come in pairs $\{x + s_1, x + s_2\}$ where $x - y = s_1 + s_2$ is an expression for the difference of the vertices as a sum of elements from S . When $x - y \in T$, the number of paths of length 2 between the vertices may be even or odd.

(b) Suppose that there exist $x, y, z \in S$ such that $x + y = z$, then the vertices $0, x, z = x + y$ are all joined by edges and form a triangle. Conversely, suppose that g_1, g_2, g_3 form a triangle in (A, S) . Then $h_1 := g_1 - g_2 \in S$, $h_2 := g_1 - g_3 \in S$ and $h_3 := g_2 - g_3 \in S$. Clearly, $h_1 + h_2 = h_3$, where all three elements belong to S .

(c) Suppose that (A, S) has diameter 2 and consider $z \notin S$. Let $g \in A$, then g and $g + z$ are not adjacent, but there exists a common neighbour h . It follows that $x := g + h \in S$ and $y := g + z + h \in S$, hence there exist $x, y \in S$ such that $x + y = z$. Vice versa, assume that for all $z \notin \{S \cup \{0\}\}$ there exist $x, y \in S$ such that $x + y = z$, and let g_1 and g_2 be two non-adjacent vertices. Since $z = g_1 + g_2 \notin \{S \cup \{0\}\}$, there exists $x, y \in S$ such that $x + y = z$. It follows that $g_1 + y$ is a common neighbour of g_1 and g_2 : $g_1 + g_1 + y = y \in S$ and $g_2 + g_1 + y = z + y = x \in S$.

(d) The graph (A, S) is $K_{2,3}$ -free if any two non-adjacent vertices have at most two common neighbours. First, let g_1, g_2 be non-adjacent vertices so that $g_1 + g_2 = z \notin \{T \cup \{0\}\}$. From the proof of the first claim, every pair $x, y \in S$ with $x + y = z$ gives rise to two common neighbours of g_1 and g_2 . Thus, if (A, S) is $K_{2,3}$ -free then each $z \notin \{S \cup T \cup \{0\}\}$ has at most one expression as a sum $x + y$ where $x, y \in S$. If $z \in T$ then there exists at least one $x \in S$ such that $x + x = z$. If z could be expressed as a sum of distinct elements of S , then there would be at least three common neighbours of 0 and z in (A, S) ; hence no element of T can be written as a sum of distinct elements of S . Finally, if $z \neq 0$ then there can be at most two elements $x, y \in S$ such that $x + x = y + y = z$ otherwise, (A, S) would contain a copy of $K_{2,3}$.

If A is a group of odd order, then $|T| = |S|$. Since S is inverse-closed, S and T intersect whenever S contains an element of order 3; such Cayley graphs are never triangle-free. If A has even order, then the number of paths from 0 to an element of T must be odd because $x + x = y + y$ implies that $x = y$, because multiplication by 2 is an automorphism of the group. If A is an elementary abelian 2-group, then $T = \{0\}$.

Proposition 4 *Suppose that A is an abelian group of order n , and S is an inverse-closed subset of A of order k , such that (A, S) is triangle-free, $K_{2,3}$ -free and has diameter 2. If A has order n which is coprime to 6, then $2n - 1$ is the square of an integer. If A is an elementary 2-group of order $n = 2^d$ then $2^{d+1} = k^2 + k + 2$.*

Proof. Suppose first that A has order coprime to 6. By [Lemma 1](#), each $z \in T$ has a unique expression $z = x + x$ for $x \in S$ and for each element $z \notin \{S \cup T \cup \{0\}\}$ there exists a unique subset $\{x, y\} \subseteq S$ such that $z = x + y$. Since S is inverse-closed, the multiset $S^2 = \{x + y \mid x, y \in S\}$ contains 0 precisely k times, and since (A, S) is triangle-free, no element of S occurs in S^2 . Hence,

$$|S|^2 = k + |T| + 2|G \setminus \{S \cup T \cup \{0\}\}|,$$

which implies that $k^2 = 2k + 2(n - 2k - 1)$. This is easily rewritten as $2n - 1 = (k + 1)^2$, yielding the conclusion.

If A is an elementary abelian 2-group of order 2^d then each element $z \notin S \cup \{0\}$ admits a unique expression $z = x + y$ for $x, y \in S$. Hence,

$$|S|^2 = k + 2|G \setminus \{S \cup \{0\}\}|,$$

which implies that $k^2 = k + 2(2^d - k - 1)$. This is easily rewritten as $2^{d+1} = k^2 + k + 2$ completing the proof.

[Proposition 4](#) proves that many elementary abelian groups of odd order do not support a Cayley graph which is triangle-free, $K_{2,3}$ -free and has diameter 2. The prime power orders less than 200 at which such a graph is not ruled out are 5, 13, 25, 41, 61, 113, 181. Solutions exist for the first two orders, but for none of the others by the computations described in [Section 3](#).

Corollary 1. *If $(\mathbb{Z}_2^d, S)$ is triangle-free, $K_{2,3}$ -free and has diameter 2, then $(\mathbb{Z}_2^d, S)$ is a $\text{srg}(2^d, |S|, 0, 2)$.*

The main result in this section is a proof that [Conjecture 1](#) holds for Cayley graphs on elementary abelian 2-groups. For $G = \mathbb{Z}_2^2$ which has only four vertices, the condition to be $K_{2,3}$ -free is automatically met, and it is easy to see that the only diameter 2 triangle-free graph in this case is $K_{2,2}$. This can also be described as the Cayley graph $(\mathbb{Z}_2^2, S)$ with $S = \{10, 01\}$.

So from now on, we only consider the case $n \geq 3$. We will show that there is a unique example of a triangle-free, $K_{2,3}$ -free Cayley graph on $\mathbb{Z}_2^d$. This example is the well-known *Clebsch graph*. This is a $\text{srg}(16, 5, 0, 2)$ and the unique strongly regular graph with these parameters, which can also be described as the Cayley graph $(\mathbb{Z}_2^4, S)$ with $S = \{1000, 0100, 0010, 0001, 1111\}$.

Theorem 5. *Let $d \geq 3$. If $(\mathbb{Z}_2^d, S)$ is triangle-free, $K_{2,3}$ -free and has diameter 2, then $(\mathbb{Z}_2^d, S)$ is the Clebsch graph $\text{srg}(16, 5, 0, 2)$.*

Proof. By [Lemma 1](#), we need to find $S \subseteq \mathbb{Z}_2^d$ such that (a) $x + y = z$ has no solution with $x, y, z \in S$; (b) for all $z \notin \{S \cup \{0\}\}$, there is a unique pair $x, y \in S$ such that $x + y = z$.

By [Proposition 4](#), these conditions imply that $k^2 + k + 2 - 2^{d+1} = 0$. This is a quadratic equation in k , with roots equal to $(-1 \pm \sqrt{2^{d+3} - 7})/2$. Since k is the size of a set, $2^{d+3} - 7$ must be the square of an integer. This is the Ramanujan–Nagell

equation, and Nagell [19] showed that the only values of m for which $2^m - 7$ is a square are 3, 4, 5, 7, 15.

Since $d + 3 \geq 6$, we find $d = 4$ and $d = 12$ are the only admissible values. If $d = 4$, then $|S| = 5$ and the parameters of the strongly regular graph (G, S) are $(16, 5, 0, 2)$. We conclude that (G, S) is the Clebsch graph.

If $d = 12$, then $|S| = 90$. The corresponding graph would be a $\text{srg}(2^{12}, 90, 0, 2)$. The parameters of a $\text{srg}(v, k, \lambda, \mu)$ must satisfy the following integrality condition, expressing that the multiplicities of its eigenvalues are integers:

$$\frac{1}{2}(v-1) \pm \frac{2k + (v-1)(\lambda - \mu)}{\sqrt{(\lambda - \mu)^2 + 4(k - \mu)}} \in \mathbb{Z}.$$

Since $\lambda = 0$ and $\mu = 2$ this condition reduces to $\frac{|S| - 2^n + 1}{\sqrt{|S| - 1}} \in \mathbb{Z}$. Since $\sqrt{90 - 1}$ is not an integer, no such graph exists.

We conclude this section with a comment on the connections between difference sets and strongly regular graphs. A (v, k, λ) -*difference set* in a finite group G of order v is a subset $D \subseteq G$ of size k such that every non-identity element $g \in G$ admits a fixed number λ of expressions $d_i - d_j = g$ (including the elements of D). A difference set is non-trivial if it is not the whole group, or the whole group excluding a single element. A difference set in an elementary abelian 2-group not containing 0 corresponds to a Cayley strongly regular graph with $\lambda = \mu$. A difference set may be translated by setting $D' = D + g$; by taking $g \in D$, and discarding the identity element, one obtains a reduced strongly regular graph with $\lambda = \mu - 2$.

By a Theorem of Mann [17], a non-trivial difference set in an elementary abelian 2-group necessarily has parameters $(2^{2n+2}, 2^{2n+1} - 2^n, 2^{2n} - 2^n)$ for some integer n . Such a difference set corresponds to a triangle-free strongly regular graph if and only if $n = 1$, in which case the reduced strongly regular graph has parameters $(16, 5, 0, 2)$, yielding an alternate (and less direct) proof of Theorem 5. The example on four vertices arises from a trivial $(4, 3, 2)$ difference set.

5 Possible Generalisation

This section considers the natural question: can $K_{2,t}$ in Conjecture 1 be replaced by $K_{s,t}$?

If v is a vertex in a graph G , then let G' be any graph obtained from G by replacing v by an independent set, each vertex of which has the same neighbourhood as v . Then G' is said to be obtained from G by *blowing-up* v . This operation maintains triangle-freeness and diameter 2.

Open Problem 6 *Is it true that for all integers $s, t \geq 1$, there exists n_0 such that for every triangle-free $K_{s,t}$ -free diameter-2 graph G , there exists a triangle-free $K_{s,t}$ -free diameter-2 graph G_0 with at most n_0 vertices and there exists an independent set*

I in G_0 of vertices with degree at most $s - 1$, such that G is obtained from G_0 by blowing-up each vertex in I ?

Distinct vertices v, w in a graph G are *twins* if $N_G(v) = N_G(w)$. Note that twins are not adjacent. A graph G is *twin-free* if no two vertices in G are twins.

Open Problem 7 *Is it true that for all integers $s, t \geq 1$, there exists n_0 such that if G is a triangle-free twin-free $K_{s,t}$ -free graph with diameter 2, then G has at most n_0 vertices?*

Since blowing-up a vertex produces twins, a positive answer to [Open Problem 6](#) implies a positive answer to [Open Problem 7](#). The converse also holds.

Proposition 8 *A positive answer to [Open Problem 7](#) implies a positive answer to [Open Problem 6](#).*

Proof. Fix $t \geq s \geq 1$. Let n_0 be as in [Open Problem 7](#). Let G be a triangle-free $K_{s,t}$ -free diameter-2 graph G . For vertices $v, w \in V(G)$, say $v \sim w$ if $N_G(v) = N_G(w)$. Then $\sim$ is an equivalence relation. Let $V_1, \dots, V_m$ be the corresponding equivalence classes. Then each V_i is an independent set. Say V_i is *small* if $|V_i| \leq t - 1$ and *large* otherwise. Let G_0 be obtained from G by identifying the vertices in each set V_i . Let G_1 be obtained from G by identifying the vertices in each large set V_i . Let L be the set of vertices in G_1 obtained by these identifications. So L is an independent set in G_1 , as otherwise $K_{t,t}$ would be a subgraph of G . Moreover, each vertex in L has degree at most $s - 1$ in G_1 , as otherwise $K_{s,t}$ would be a subgraph of G . Since G_0 and G_1 are isomorphic to subgraphs of G , G_0 and G_1 are triangle-free and $K_{s,t}$ -free. Since G has diameter 2, so does G_0 and G_1 . By construction, G_0 is twin-free. By the assumed truth of [Open Problem 7](#), $|V(G_0)| \leq n_0$. By construction, $|V(G_1)| \leq (t - 1)|V(G_0)|$. Thus $|V(G_1)| \leq (t - 1)n_0$. We have shown that G_1 is a triangle-free $K_{s,t}$ -free diameter-2 graph with at most $(t - 1)n_0$ vertices and there exists an independent set L in G_1 of vertices with degree at most $s - 1$, such that G is obtained from G_1 by blowing-up each vertex in L . Hence G satisfies [Open Problem 6](#).

In each of Conjecture 1 and Open Problems 6 and 7, the triangle-free condition cannot be dropped (unless $s = 1$ or $s, t \leq 2$). For example, wheel graphs are twin-free $K_{2,3}$ -free with diameter 2. In general, if G is any twin-free $K_{s,t}$ -free graph (regardless of the diameter), then the graph G' obtained from G by adding one dominant vertex is twin-free $K_{s+1,t+1}$ -free with diameter 2. Of course, G' has many triangles. There are examples without dominant vertices as well ($K_{2,2,n}$ is $K_{5,5}$ -free, diameter 2, and no dominant vertex).

To conclude this section, we make some elementary observations about edge-maximal graphs. Fix an integer $k \geq 3$. Let $\mathcal{H}$ be any set of graphs such that for each $H \in \mathcal{H}$, each edge in H is in a cycle of length at most k . Then every edge-maximal $\mathcal{H}$ -free graph G has diameter at most $k - 1$. Otherwise, there are vertices v, w in G with $\text{dist}_G(v, w) = k$, implying $G + vw$ contains a copy of a graph H in $\mathcal{H}$ and vw is in the copy, implying there is a vw -path in G of length at most $k - 1$, which contradicts $\text{dist}_G(v, w) = k$. Taking $\mathcal{H} = \{K_3\}$, this implies:

every edge-maximal triangle-free graph has diameter at most 2.

And taking $\mathcal{H} = \{K_3, K_{s,t}\}$, for any fixed $s, t \geq 2$, this implies:

every edge-maximal K_3 -free $K_{s,t}$ -free graph has diameter at most 3.

Open Problem 7 asks whether this diameter bound can be improved from 3 to 2 for some infinite family without twins.

6 Random Graphs

A natural avenue for disproving **Open Problem 7** would be the probabilistic method, which has yielded diverse and important results on Ramsey- and Turán-type questions. For instance, in the classical Zarankiewicz problem, essentially the strongest lower bound on the number of edges in an n -vertex $K_{s,t}$ -free graph which is valid for all values of s and t ,

$$\text{ex}(n, K_{s,t}) \geq \varepsilon n^{2 - \frac{s+t-2}{st-1}}, \quad (1)$$

is obtained using the so-called *alteration method* (see [12, Theorem 2.26]). Our aim would be to construct a triangle-free $K_{s,t}$ -free graph with diameter 2 and no twin vertices. The following discussion refers to the n -vertex random graph $G(n, m)$ which has m edges selected uniformly at random.² Let us start by noting that the *threshold* at which $G(n, m)$ has diameter 2 is asymptotically $m^* = \sqrt{\frac{1}{2}n^3 \log n}$ [18]. This can be heuristically justified by considering the expected number of pairs with no common neighbours, which is asymptotic to $\frac{1}{2}n^2 e^{-4m^2 n^{-3}}$.

Firstly, we note that one cannot hope to avoid $K_{2,t}$ -copies or $K_{3,3}$ -copies using purely probabilistic methods. To see this, we discuss the alterations method. To ‘construct’ a $K_{s,t}$ -free graph with $\varepsilon n^{2 - \frac{s+t-2}{st-1}}$ edges, start with the Erdős–Rényi random graph with $m = 2\varepsilon n^{2 - \frac{s+t-2}{st-1}}$ edges and remove an edge from each $K_{s,t}$ -copy and each triangle, losing at most a half of the edges (see, e.g. [12]). This argument fails when $m > n^{2 - \frac{s+t-2}{st-1}}$ since then the expected number of $K_{s,t}$ copies exceeds the number of edges. For $s = 2$ or $s = t = 3$ this threshold is at most $n^{3/2} < m^*$, so the obtained graphs typically do not have diameter 2. More advanced tools, such as the $K_{s,t}$ -free process discussed soon, fail for the same reason—an average edge of $G(n, m)$ is contained in many $K_{s,t}$ -copies.

Hence, we will discuss an approach to **Open Problem 7** for $s = 3$ and $t = 4$. Note that maximal triangle-free graphs are exactly minimal graphs of diameter 2. An approach that seems tailored for constructing such graphs is a *constrained* random graph process, defined as follows. Let $\mathcal{C}$ be a monotone class of graphs (that is,

² For the purpose of our discussion, this is essentially equivalent to selecting each edge independently with probability $p = m \binom{n}{2}^{-1}$, but we use the model with m random edges since it is easier to relate it to the H -free process discussed later.

closed under taking subgraphs). Let $e_1, \dots, e_{\binom{n}{2}}$ be a uniformly random ordering of the edges of K_n , with n an even integer. For $m \geq 0$, G_{m+1} is obtained from G_m by adding the edge e_{m+1} exactly if $G_m \cup \{e_{m+1}\}$ is in $\mathcal{C}$, starting from the empty n -vertex graph G_0 . When $\mathcal{C}$ is the class of H -free graphs, this process (referred to as the *H -free process*) is fairly well understood, and the implied lower bound on the extremal function of $K_{s,s}$ when $s \geq 5$ exceeds (1) by a polylogarithmic factor [5, Theorem 1.1]. Particular attention has been given to the case when H is a triangle because of its implications for the off-diagonal Ramsey number $r(3, t)$ [6, 10].

Consider the random $\mathcal{C}$ -process where $\mathcal{C}$ is the class of K_3 -free and $K_{3,4}$ -free graphs, and let G be the final graph in the process (that is, the random graph which is both K_3 - and $K_{3,4}$ -saturated). If we could show that G has diameter 2 with positive probability, then this would likely answer [Open Problem 7](#) in the negative. This is not an unreasonable event to expect for several reasons. First, both the K_3 -free process and the $K_{3,4}$ -free process typically result in a graph with at least $(1 - o(1))m^*/2$ edges, and the latter has significantly more edges (see for example, [5, Theorem 1.1] and [6, Theorem 1.1]). An even more promising feature of the H -free process is that (in the words of Bohman and Keevash [5, Corollary 1.5]) “the random H -free graph G_i is similar to the uniform random graph $G(n, i)$ with respect to small subgraph counts, with the notable exception that there are no copies of graphs containing H in G_i .” This feature is also crucial in the analysis using the differential equations method. Thus one may hope that in the random $\mathcal{C}$ -process, the co-degrees of non-adjacent vertex pairs are similar to those in the corresponding unconstrained random graph $G(n, m)$, and in particular, they are all positive at the end. A final encouraging observation is that if the distance between vertices u and v in a graph G_m is at least 3, then the edge uv can be added without creating a triangle (but it may create a $K_{3,4}$). Moreover, it suffices to find a *large* subgraph $G' \subset G$ of diameter 2. Let us state the question explicitly.

Open Problem 9 *Let $s \geq 2$ and $t \geq 3$, and let G be the (random) graph obtained from the random $\{K_3, K_{s,t}\}$ -free-process. Does G contain a subgraph G' of diameter 2, with $|V(G')| \geq f(|V(G)|)$, for some function f with $f(n) \rightarrow \infty$ as $n \rightarrow \infty$?*

Experimental investigation of [Open Problem 9](#) for $(s, t) = (2, 3)$ and $(s, t) = (3, 4)$ suggests that such a function f grows very slowly if it exists. In thousands of trials up to a few hundred vertices, no diameter 2 subgraph of order greater than 13 was encountered.

Note added in proof

After the initial release of this paper, Eberhard [8] disproved [Conjecture 1](#) in the $t = 7$ case.

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